

1 CR Remaining Gaps + Is MERN Still Right?

MERN MASTERY · 1 CR PRODUCT PREP · HARSH VARDHAN GUPTA

 **Study window:** 17 Aug 2026 – 14 Feb 2027 (Days 1–182)

 **Your slots (IST):** Mon–Fri **7:30–8:30pm** & **9:00–10:00pm** · Sat–Sun **9:00am–12:00pm, 3:00–5:30pm, 6:00–8:30pm**

 **Calendar reminders:** harsh.gupta@getreelax.com – import `calendar/mern-mastery-study.ics`

Second-pass audit of the whole `mern-mastery` folder. This file fills what was still thin after PDF #20.

Verdict up front: keep MERN. Do not restart in Java/Spring. Details in §1.

1. Is it still right to learn MERN?

Yes — for this roadmap and a ~1 CR product-company target, MERN (really TypeScript + React + Node + SQL/Redis) is still the correct primary stack.

If you switch to Java/Spring now	If you stay on MERN
Lose 2–3 months rewriting backend + projects	Projects (TaskFlow, ShopLite) stay interview-ready
DSA language change mid-prep	DSA in TypeScript is accepted at Meta, Uber, Atlassian, Adobe, Salesforce, most Microsoft, many Google FE
You compete as a junior Java hire	You compete as a fullstack product hire – the role that actually pays 80 L–1.2 Cr+ for your profile

What “MERN” means at 1 CR (do not stay on tutorial MERN):

Keep	Upgrade	Do not chase
React + TypeScript	Next.js, testing, a11y, Web Vitals	New JS framework every month
Node + Express	REST + one GraphQL lite (Meta)	Spring Boot rewrite
Mongo	PostgreSQL as default + Redis	Mongo-only thinking
JWT	OAuth, cookies, OWASP (already in PDF #8)	Kubernetes-as-identity

When MERN is *not* enough (rare for your target):

- Google **L5 backend / infra** or Amazon **SDE3 platform** teams sometimes want Java/C++/Go. Treat that as Phase 9 *if* a recruiter says the team is Java-only. Do not pre-empt it.
- DSA: if an OA portal has no JS, use **Python** for that one test (syntax in a weekend). Do not rebuild the syllabus.

Language rule: code all LeetCode in **TypeScript**. Keep a 20-line Python cheat sheet for OA portals that lack JS.

2. What the folder already covers (do not redo)

Area	Where	1 CR enough?
JS / TS / React / Next	PDFs #1-6	Yes
Node, auth, OWASP, Docker, Jest/RTL	PDFs #7-8	Yes
SQL + Mongo + Redis + indexes	PDF #9	Theory yes; timed SQL was thin → §4
Networking / HTTP	PDF #10b	Yes
DSA patterns + hard + company tags	PDFs #4, #12, #14, #20	Yes if you <i>time</i> them
LLD + backend MC + FE autocomplete	PDFs #13, #15, #20	Yes
Product HLD (Feed, Chat, Uber, YouTube, Drive, Typeahead)	PDFs #10, #20	Yes
Behavioral / LPs	PDF #16, #20 §8	Yes if stories have <i>your</i> metrics
Question-type map + company loops	PDF #20	Yes
Optional months 7-9	PDF #20 §11	Yes

Completing the PDFs is **necessary, not sufficient**. Offers come from **timed mocks + OA pass + referral + a team that is hiring**. This file adds the remaining *content* gaps.

3. Online Assessment (OA) – you never reach the loop without this

Amazon, Uber, many Microsoft, and some Meta contractor screens start with **HackerRank / CodeSignal / Hackerrank OA**, not a human.

Rules

Platform	Format	Pass bar
Amazon OA	2 DSA (medium) + work-style survey	Both DSA in time; LP-flavored survey consistent
CodeSignal (Uber / some startups)	4 questions, 70 min, score /850	Aim 750+
HackerRank (Adobe, Salesforce, many)	2-3 DSA, 60-90 min	All hidden tests green
Meta screen	2 mediums, 45 min (human or CoderPad)	Both working

Practice (Phase 8A and any weekend from Month 4):

1. Create accounts: LeetCode, CodeSignal, HackerRank.
2. Do **one** 70-min CodeSignal practice (arcade / certified).
3. Amazon-style: 2 mediums in 70 min, no IDE plugins, no copy-paste from notes.
4. Never leave a question blank — brute force + comment “optimize if time”.

OA week (insert before you apply, Day 169–171):

- Day 169: 2 timed mediums, HackerRank editor only
- Day 170: CodeSignal-style 4-question set (or 3 mediums + 1 easy)
- Day 171: Redo yesterday’s fails blind

Python emergency sheet (only if the portal has no JS):

```
from collections import defaultdict, deque, Counter
import heapq
# list, set, dict; heapq.heappush/heappop
# for i, x in enumerate(arr):
```

4. Timed SQL (Amazon / Microsoft / Adobe)

Theory is in PDF #6. Interviews ask you to **write the query in 10 minutes**.

Do these **8** (LeetCode SQL). Time: 12 min each. Phase 4 weekends + Phase 7G.

#	Problem	Pattern
1	<u>1757 Recyclable</u>	WHERE
2	<u>584 Find Customer Referee</u>	NULL
3	<u>175 Combine Two Tables</u>	LEFT JOIN
4	<u>181 Over 70</u>	Self join
5	<u>178 Rank Scores</u>	Window DENSE_RANK
6	<u>184 Dept Highest Salary</u>	GROUP BY + join
7	<u>180 Consecutive Numbers</u>	Window / self join
8	<u>185 Dept Top Three</u>	DENSE_RANK partition

Say this: “I’ll write the join first, then the window, then filter in an outer query.”

Index talk: “I’d put **(dept_id, salary)** if this runs on 10M rows.”

5. Browser internals (Meta FE / Google FE / Atlassian)

PDF #0b covers HTML/CSS. 1 CR frontend rounds ask how the browser turns URL → pixels.

Critical path (draw this in 3 minutes)

```

DNS → TCP → TLS → HTTP
  → HTML parse → DOM
  → CSS parse → CSSOM
  → DOM + CSSOM → Render tree
  → Layout (reflow) → Paint → Composite
JS can block parse (no defer/async). Large images delay LCP.

```

Interview lines:

- **Reflow vs repaint:** geometry change vs color/visibility. Batch DOM writes.
- **LCP:** largest image/text in viewport. Fix: size attributes, CDN, don’t wait on JS for hero.
- **INP / FID:** main-thread JS. Split long tasks, debounce input.
- **CLS:** reserved image/ad size. No late font swap without **size-adjust**.
- **Event loop in browser:** same microtask/macrotask as PDF #1; plus rAF before paint.

One drill (Day 166 extra 30 min): open ShopLite, name the LCP element, and one change that would cut it.

6. Accessibility (Atlassian / Google FE)

Frontend MC without a11y is a soft no at Atlassian.

Minimum on every widget:

- Inputs have `<label>` (or `aria-label`)
- Keyboard: Tab, Enter, Escape, arrows for lists
- `role="listbox"` + `aria-activedescendant` on autocomplete (PDF #15 §13)
- Focus visible; don't `outline: none` without a replacement
- Color is not the only error signal
- `prefers-reduced-motion`

2 questions they ask:

1. How do you test a11y? → keyboard-only pass + axe/lighthouse + one screen-reader pass on the happy path.
2. Why not `div` with `onClick`? → no keyboard, no role, no name. Use `button` .

7. GraphQL lite (Meta – do not build a new stack)

Meta FE often asks “REST vs GraphQL” and “how would you fetch a feed.”

Say:

- GraphQL: one `/graphql` POST, client asks for fields. Good when many clients need different shapes (feed + stories + ads).
- REST: simpler cache at HTTP layer (`GET /posts/1`), easier CDN.
- N+1: GraphQL resolvers can explode — **DataLoader** batches `userId → user` .
- Auth: still on the server; do not trust the query to hide fields (authorization in resolver).

Do not rewrite TaskFlow in GraphQL. Draw a 6-box diagram once (client → GraphQL gateway → post service / user service / DataLoader → DBs).

8. India ~1 CR offer math (so you do not under-ask)

1 CR usually means annualized TC, not 1 Cr cash in year 1.

Component	Typical 1 Cr Google L4 / Meta E4 / Amazon SDE2 (India, 2025-26 ballpark)
Base	35-50 L
Bonus / joining	5-20 L (joining often year-1 only)
Stock (RSU, 4-year vest)	25-50 L / year <i>if you annualize grant</i> ÷ 4
Year-1 cash	Often 55-75 L , not 1 Cr in the bank

Negotiate: level first, then signing + year-1 cash, then stock.

Never accept a level downgrade to “hit 1 Cr” on paper with funny ESOP math.

Sources: Levels.fyi (India + level), two humans at the company, recruiter range in writing.

Full scripts: PDF #17 Tier 5 — use 1 Cr TC as the ask, then unpack components.

9. Referral + apply ops (content was thin)

Week	Action
From Month 4	LinkedIn: 1 recruiter + 1 engineer / week at target companies. Ask for referral , not “can we chat 30 min?”
Resume	1 page, metrics, TaskFlow + ShopLite architecture links
Apply	Referral same week as OA practice, not after you “feel ready”
Tracker	Company, role, referral name, OA date, loop date (month-06 already has a table)
Volume	8 applications in Phase 8E is a minimum . 1 CR needs ~15-25 serious shots over 8 weeks

Cold apply without referral at Google/Meta is a lottery. Referral is part of the syllabus.

10. When to study this file

Section	When
§1 MERN decision	Read once, Day 1 and Day 113
§3 OA	Month 4 weekends + Days 169-171
§4 SQL	Month 4 Sunday + Day 165
§5 Browser	Day 166 (with FE MC)
§6 A11y	Every frontend MC
§7 GraphQL	Day 167 if Meta is a target
§8 Comp	Day 182 + offer week
§9 Referrals	Month 4 onward

11. Honest 1 CR scorecard (folder + your work)

The folder can get you to the **hire bar**. It cannot sit the interview for you.

You did the folder	You still fail if
220 LC, 10 product HLDs, 2 FE widgets, 10 STAR	You never timed an OA
All PDFs highlighted	You cannot defend TaskFlow for 20 min without notes
Company loops memorized	Zero referrals, 3 cold applies
1 CR ask rehearsed	You freeze when they change the HLD constraint

Minimum extra hours this file adds: ~20–25 (OA + 8 SQL + browser drill + a11y on widgets). That is the last content gap.

Still not in any PDF (by design): a real job, a real team opening, and luck on interviewer. Plan Phase 9 if three full-loop mocks are not “hire.”